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(54) **APPARATUS AND METHOD FOR TREATING INTERSTITIAL CYSTITIS**

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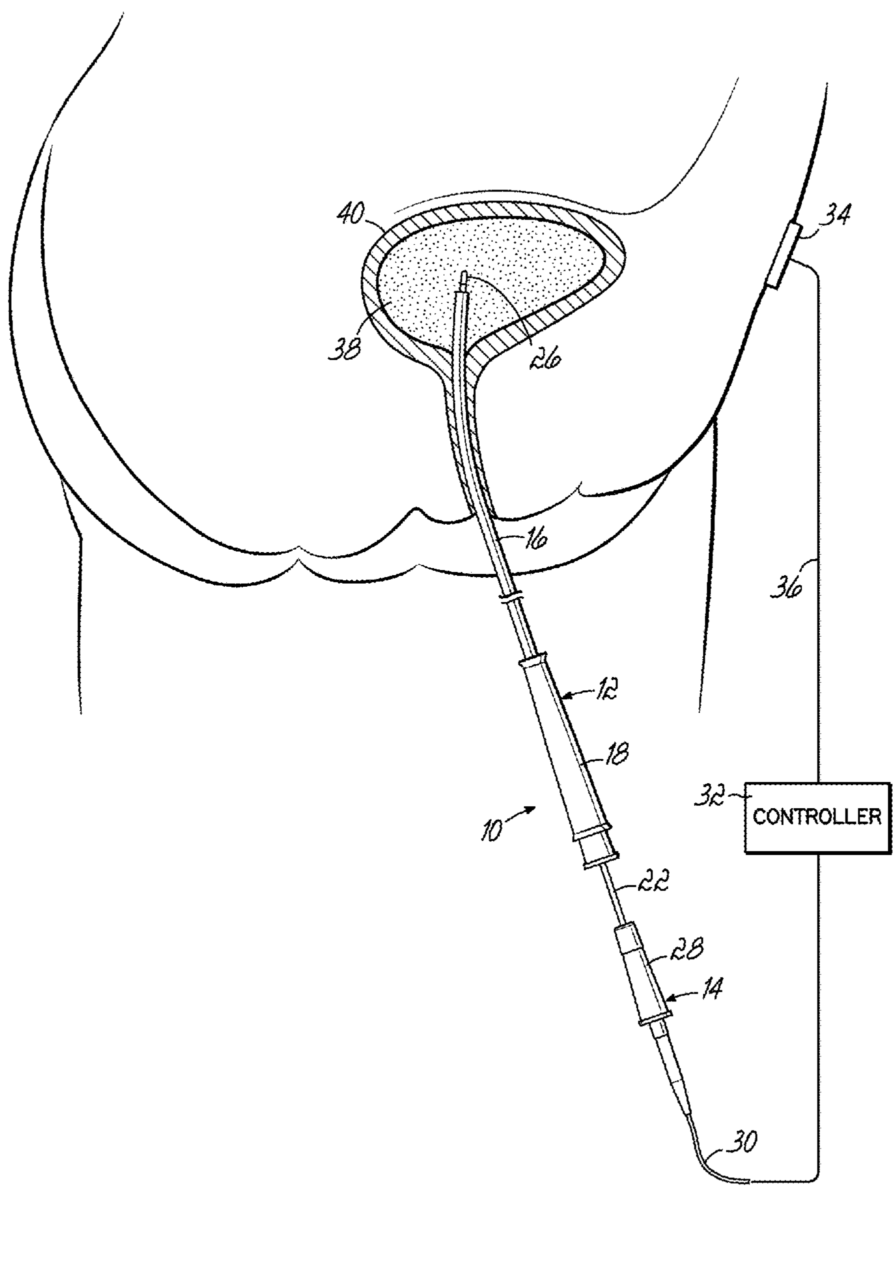
(57) **ABSTRACT**

A method for treating interstitial cystitis of a patient is disclosed. The method includes providing a catheter with a lumen and an elongate rod disposed within the lumen and routing the lumen through a urethra such that a distal end of the elongate rod is disposed in an interior of a bladder. Then connecting the elongate rod to a controller adapted to generate current and supplying a current at a first amperage from the controller to the distal end of the elongate rod. Applying the current for a predetermined time so as to disrupt a biofilm on or proximate to an interior wall of the bladder.



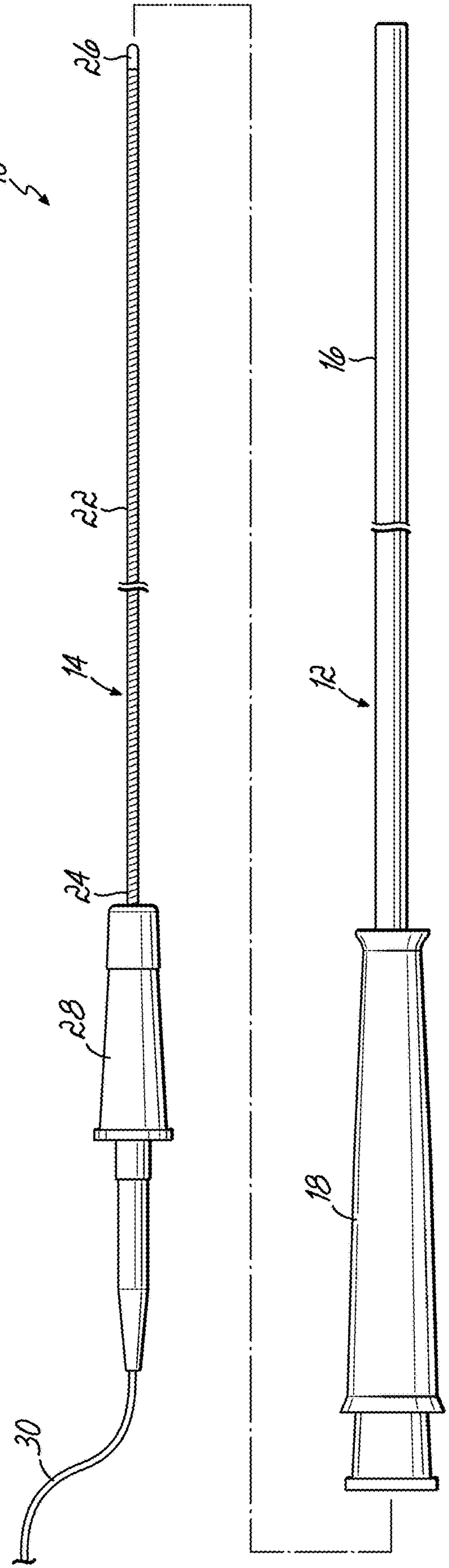


FIG. 1A

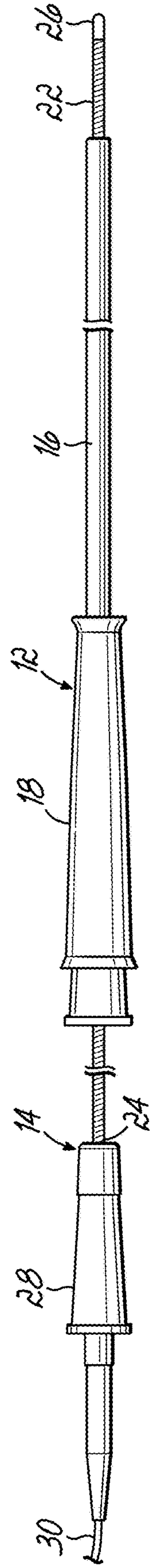


FIG. 1B

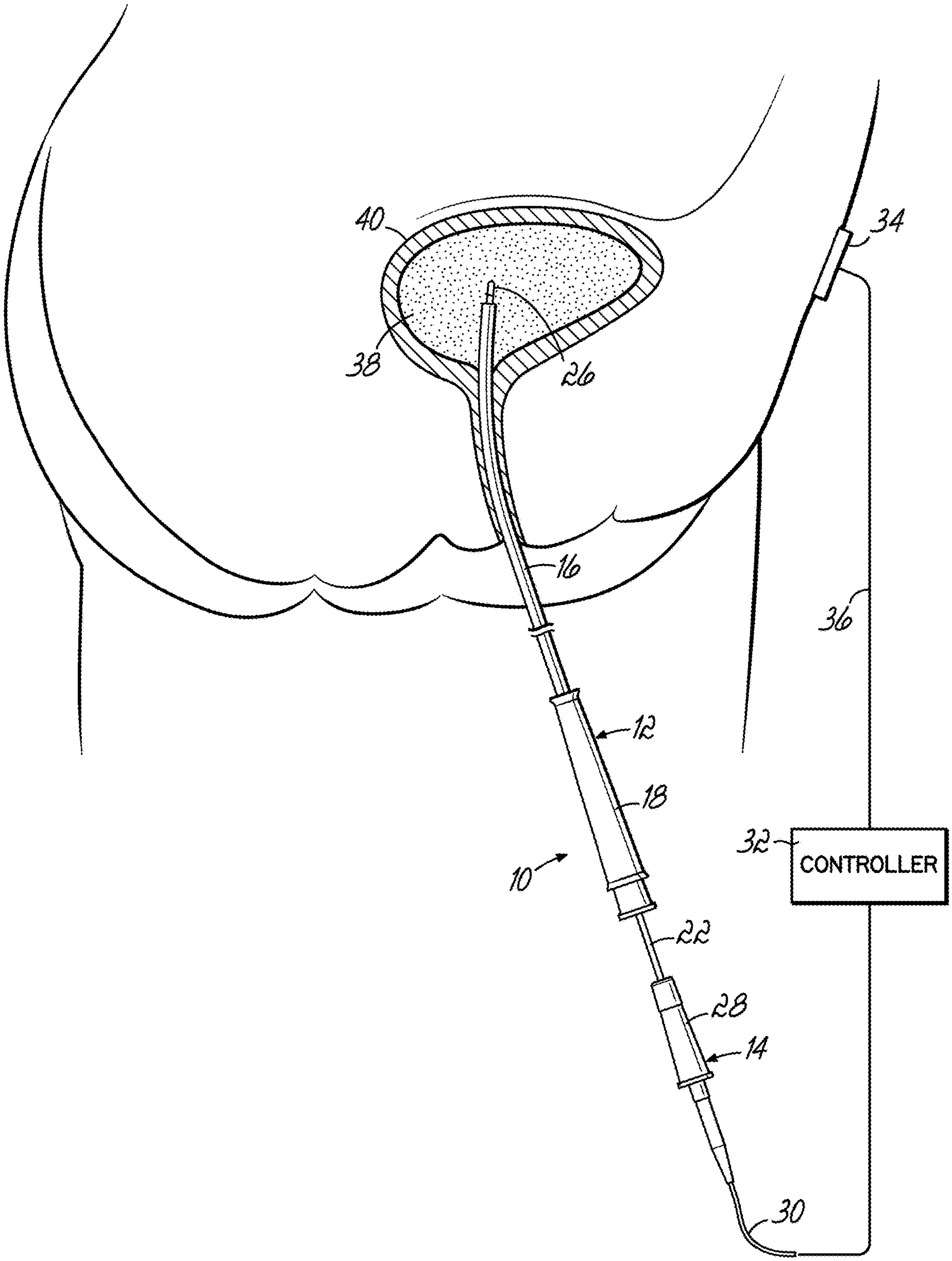


FIG. 2

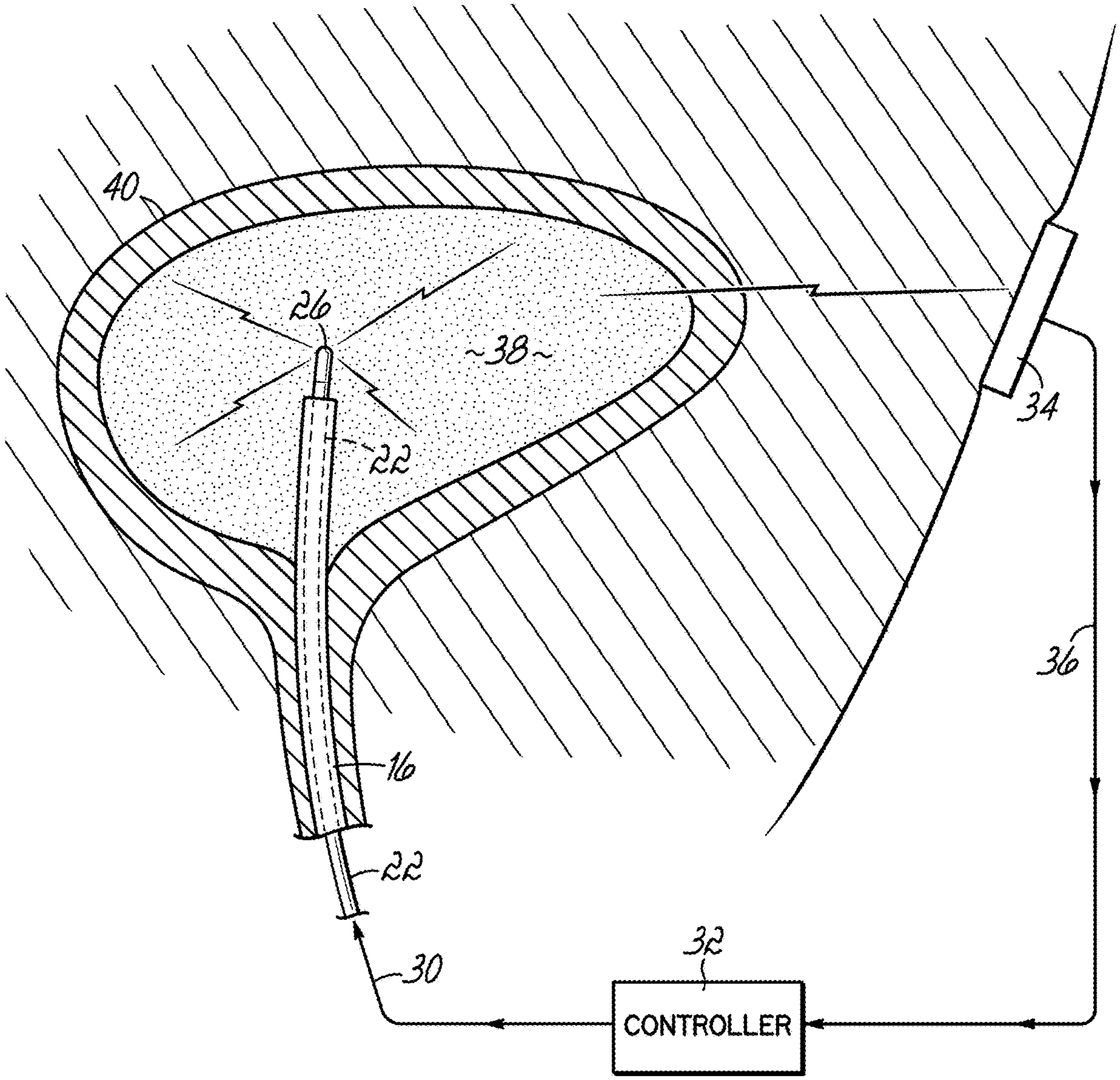


FIG. 3

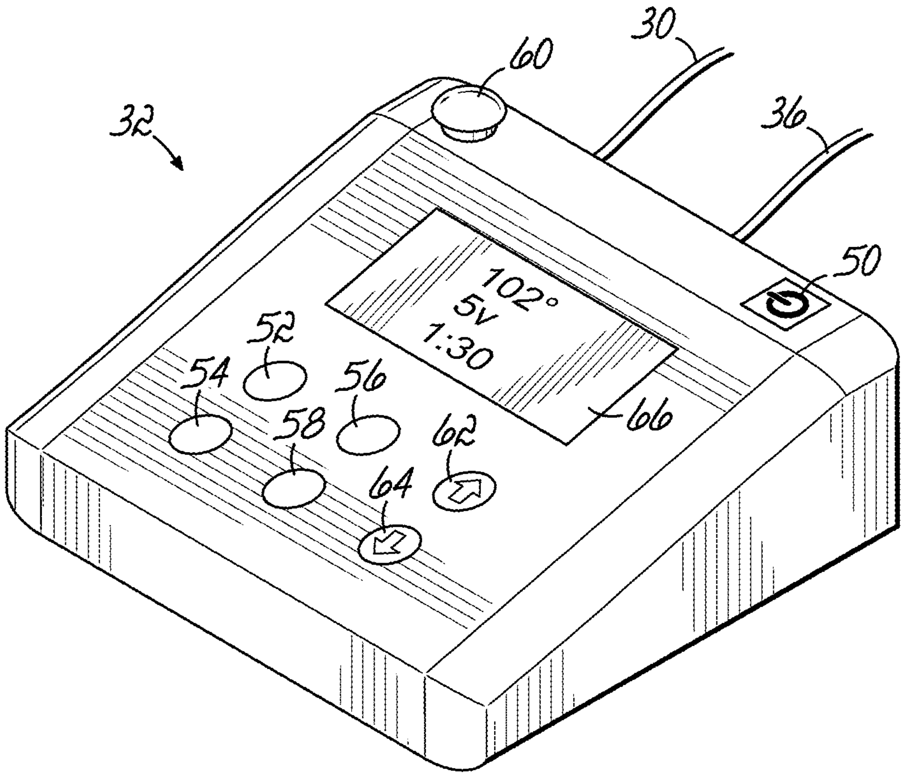


FIG. 4

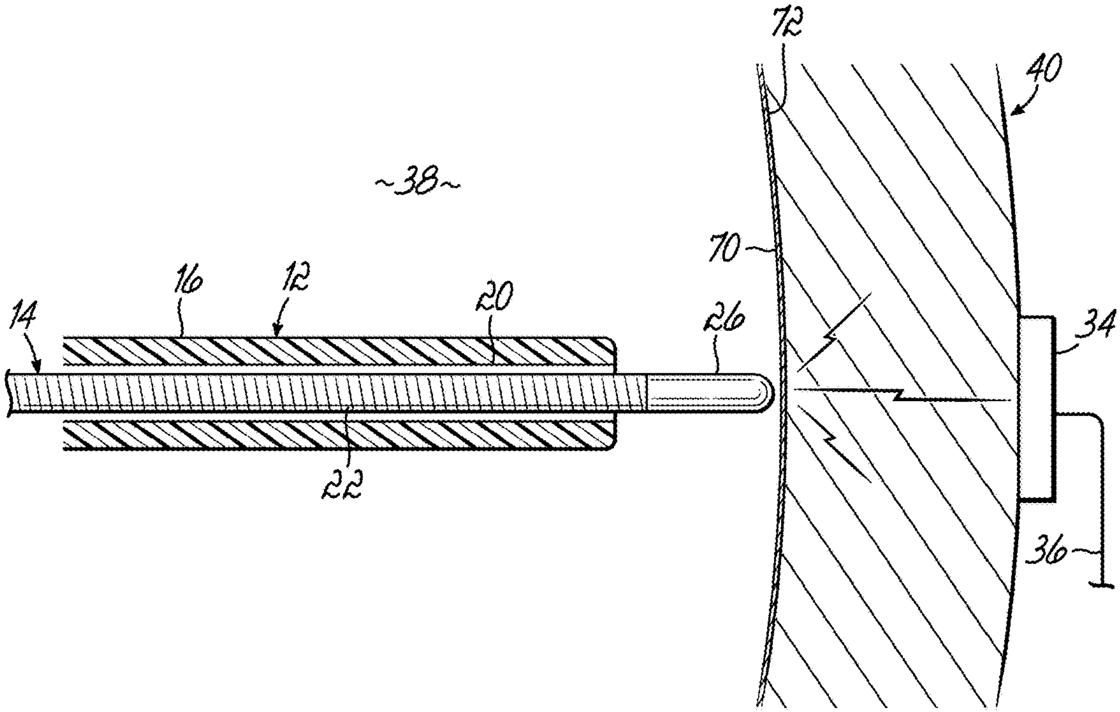

72
40
~38~
70
16
12
20
14
26
34
22
36

FIG. 5

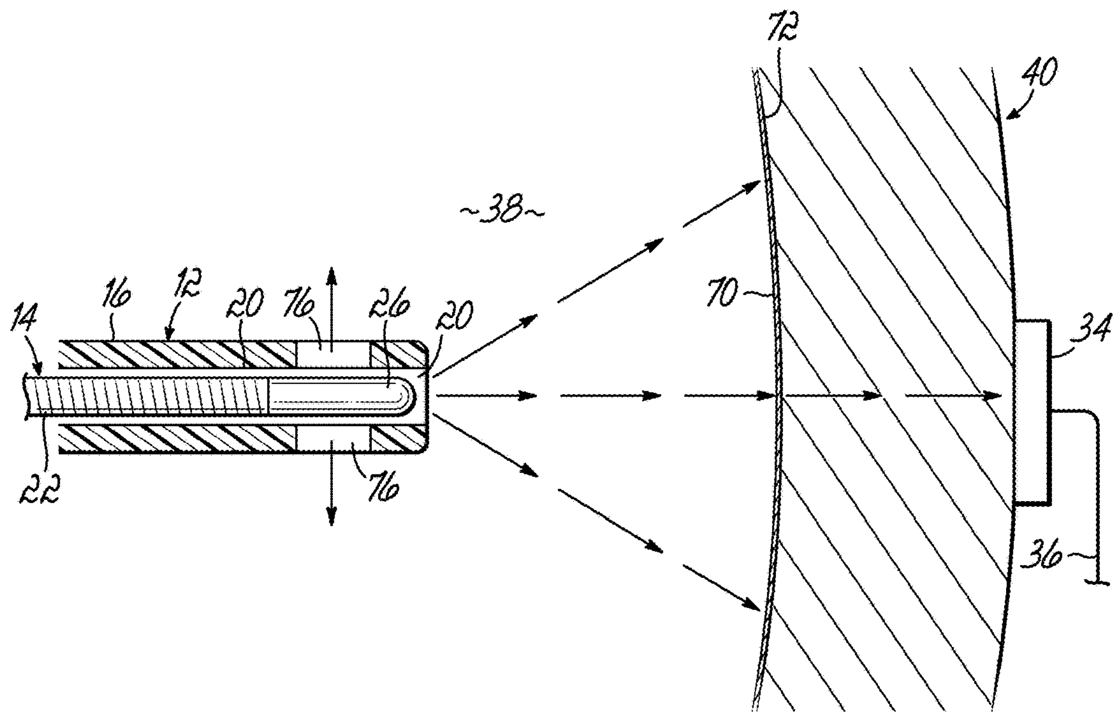

72
40
~38~
16
12
20
76
26
20
14
70
34
22
76
36

FIG. 6

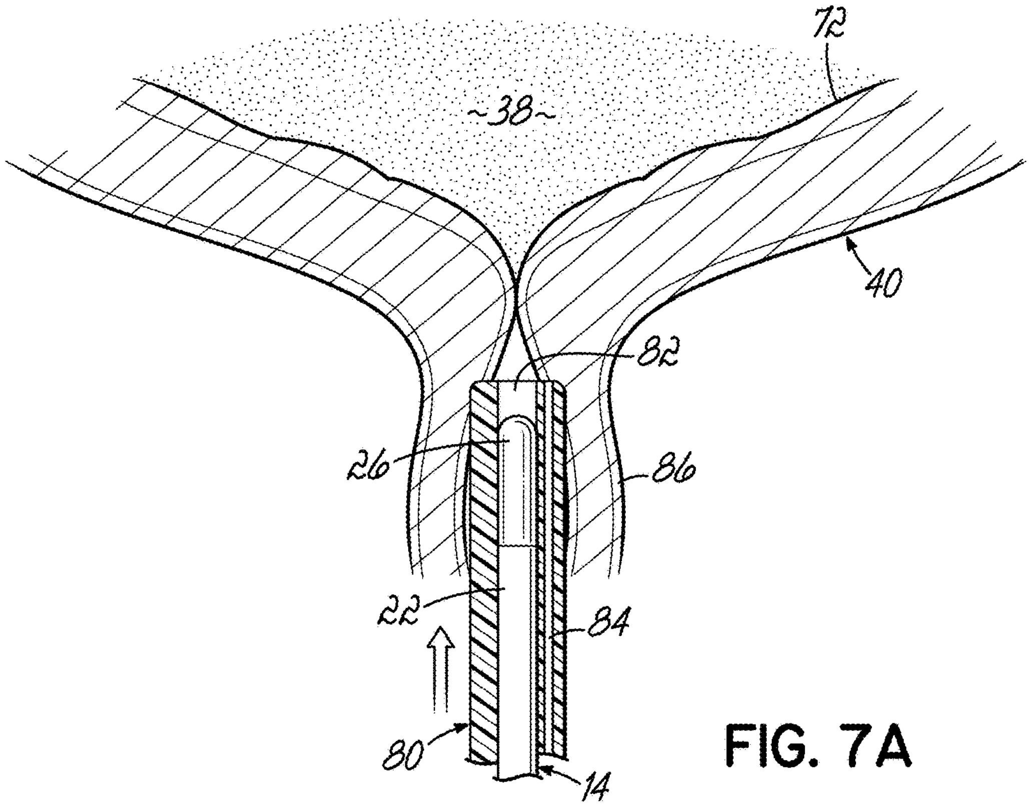


FIG. 7A

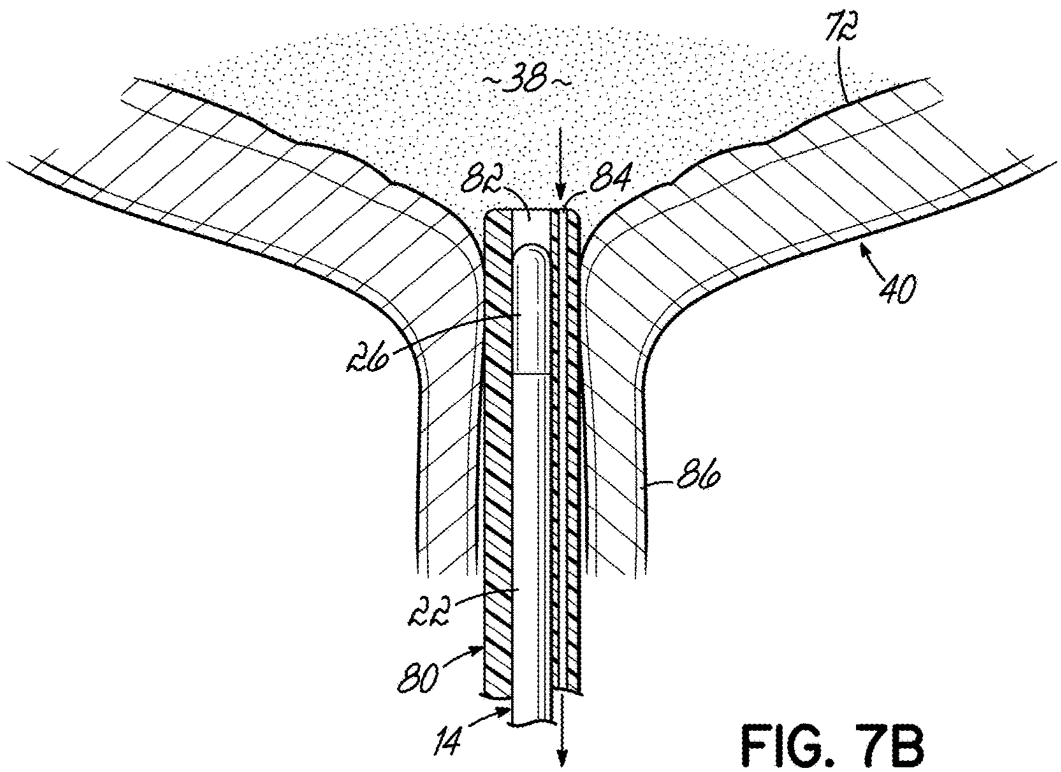


FIG. 7B

APPARATUS AND METHOD FOR TREATING INTERSTITIAL CYSTITIS

CROSS REFERENCE TO RELATED APPLICATION

[0001] This application claims the benefit of U.S. Provisional Patent Application Ser. No. 63/557,160, filed Feb. 23, 2024, the disclosure of which is incorporated by reference herein in its entirety.

TECHNICAL FIELD

[0002] The invention relates generally to an apparatus and method for treating a condition of the bladder such as interstitial cystitis.

BACKGROUND

[0003] Interstitial cystitis (IC) is an inflamed or irritated bladder wall. It may also be referred to as painful bladder syndrome or frequency-urgency-dysuria syndrome. IC may lead to scarring and stiffening of the bladder. As such, the bladder may not be capable of holding as much urine as it did in the past. IC is a chronic disorder and the cause of IC is unknown. Symptoms of IC may include frequent urination, urgency with urination, pain and tenderness around the bladder and pelvis, and pain during sex.

[0004] Currently, there is no cure for IC. In addition, IC may be difficult to treat. The various treatments focus primarily on easing the symptoms of IC. For example, treatments may include bladder enlargement, bladder wash, medicine, transcutaneous electrical nerve stimulation (TENS), bladder training, and surgery.

[0005] What is needed, however, is a method to effectively treat and resolve IC instead of merely focusing on treating the symptoms of IC.

SUMMARY OF THE INVENTION

[0006] A method for treating interstitial cystitis of a patient includes generating a current at a first amperage in an interior of a bladder and applying that current for a predetermined time so as to disrupt a biofilm on or proximate to an interior wall of the bladder. In one aspect, the predetermined time may be 1-6 minutes or 2-5 minutes.

[0007] In one aspect, the method may further include after generating the current at the first amperage, increasing the current to a second amperage.

[0008] In one aspect, the method may further include placing a first end of an elongate rod inside an interior of the bladder and emitting the current from the first end of the elongate rod. The method may further include operatively connecting the first end of the metal probe to a controller and changing the current from the first amperage via the controller.

[0009] In one aspect, the elongate rod may be disposed within a lumen routed through the urethra and into the bladder.

[0010] In one aspect, during the treatment method, the patient may be in a supine position, in a prone position, or standing during the treatment method.

[0011] In another embodiment, a method for treating interstitial cystitis of a patient includes providing a catheter with a lumen and an elongate rod disposed within the lumen and routing the lumen through a urethra such that a distal end of the elongate rod is disposed in an interior of a bladder. The elongate rod is then connected to a controller adapted to generate current. A current is then supplied at a first amperage from the controller to the distal end of the elongate rod. The current is applied for a predetermined time so as to disrupt a biofilm on or proximate to an interior wall of the bladder. In one aspect, the predetermined time may be 1-6 minutes or 2-5 minutes.

[0012] In one aspect, the method may further include after generating the current at the first amperage, increasing the current to a second amperage.

[0013] In one aspect, during the treatment method, the patient may be in a supine position, in a prone position, or standing during the treatment method.

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate one or more embodiments of the invention and, together with a general description of the invention given above, and the detailed description given below, serve to explain the invention.

[0015] FIG. **1**A is disassembled view of a treatment device including a flexible rod and a urinary catheter according to an embodiment of the invention.

[0016] FIG. **1**B is the assembled treatment device of FIG. **1**A.

[0017] FIG. **2** shows the treatment device positioned to treat a condition within the bladder.

[0018] FIG. **3** is an up close view of the treatment device functioning to treat a condition within the bladder.

[0019] FIG. **4** is a perspective view of a controller adapted to control the treatment device during a treatment.

[0020] FIG. **5** is a close up, cross-sectional view of the treatment device inside the bladder with the end of the flexible rob located external to the lumen of the urinary catheter according to an embodiment of the disclosure.

[0021] FIG. **6** is a close up, cross-sectional view of the treatment device inside the bladder with the end of the flexible rob located internal to the lumen of the urinary catheter according to another embodiment of the disclosure.

[0022] FIG. **7**A is a cross-section view of a treatment device according to an embodiment of the disclosure positioned in the urethra near the bladder where lumen has a side port.

[0023] FIG. **7**B is a cross-sectional view of the treatment device of FIG. **7**A with its end positioned inside the bladder and contacting urine therein.

DETAILED DESCRIPTION OF THE INVENTION

[0024] One theory for what causes IC involves a biofilm that is growing on the inside wall of the bladder. Bacteria growing on any surface that have access to moisture will invariably grow polysaccharide/protein biofilms. While starting as a typical urinary tract infection, if the particular strain of bacteria happens to be aggressive biofilm formers there would be a good chance of biofilm formation on the inner bladder wall. The biofilm is not an infection, but is instead a colonization that allows bacteria to produce exotoxins that lead to chronic pain and inflammation. That inflammation is known as IC.

[0025] It is well known that a low voltage (AC or DC) current passed through a biofilm will result in a devastating vaporization of that biofilm within just a few seconds to minutes.

[0026] With reference to FIGS. **1**A and **1**B, a treatment device **10** is shown according to an embodiment of the disclosure. The treatment device **10** includes a urinary catheter **12** and an energy delivery device **14**. The urinary catheter **12** shown is an intermediate urinary catheter, but any suitable catheter may be used. The urinary catheter **12** includes a lumen **16** and a handle **18**. The lumen **16** includes a central passageway **20** (FIG. **5**) extending the length of the urinary catheter **12**. As is typical with urinary catheters, the lumen **16** is flexible so it may be passed through the urethra and into the bladder. The energy delivery device **14** includes an elongate rod **22** adapted to fit within the central passageway **20** of the lumen. In an embodiment, the elongate rod **22** forms a water-tight seal with the central passageway **20** of the lumen **16**. In one aspect, the portion of the metal rod protruding from the catheter may be sealed within the proximal end of the catheter with a silicon sealant. The elongate rod **22** may be flexible so that the elongate rod **22** may be passed through the urethra and into the bladder. The elongate rod **22** may be made from brass, copper, or any suitable electrically-conducting material. The elongate rod **22** includes a proximal end **24** and a distal end **26**. The proximal end **24** is operatively coupled to a handle **28**. A conductive wire **30** is operatively coupled to the proximal end **24** of the elongate rod **22**. While the conductive wire **30** is shown extending through the handle **28** to reach the proximal end **24** of the elongate rod **22**, other configurations may be used to connect the conductive wire **30** to the elongate rod **22**. In FIG. **1**B, the elongate rod **22** is positioned within the central passageway **20** of the lumen **16**.

[0027] With the elongate rod **22** positioned within the central passageway **20**, the lumen **16** may then be inserted into and passed through the urethra and into the bladder as shown in FIGS. **2** and **3**. The conductive wire **30** may be operatively connected to a controller **32**. A grounding pad **34** may be secured to the patient's abdomen or suprapubis area. A conductive wire **36** may by operatively coupled to the grounding pad **34** and extend to and operatively couple to the controller **32**.

[0028] As schematically shown in FIG. **3**, the controller **32** is adapted to generate AC or DC current that passes through conductive wire **30**, through elongate rod **22**, through urine **38** in bladder **40**, through grounding pad **34**, through conductive wire **36** and back to the controller **32**.

[0029] One embodiment of the controller **32** is shown in FIG. **4**. The controller has an on/off button **50** and various other control buttons **52-62**. Buttons **52**, **54** may be used to increase or decrease the AC or DC current flowing through the elongate rod **22**. Buttons **56**, **58** may be used to set a timer for how long the AC or DC current will flow. The controller **32** may also include an emergency shut down button **60**, which when pushed will stop the flow of AC or DC current. Buttons **62**, **64** may be used to adjust other parameters during the treatment. The controller **32** may include further buttons to adjust, stop, or start a functionality of the controller **32**. As used herein, the term button may include a push button, a switch, a toggle, a slide, a rotary knob, or any other device suitable for adjusting a parameter of the controller **32**. The controller may further include a screen **66** which is configured to display information such as temperature of the elongate rod **22**, volts, and time remaining, for example. Other information or parameters may also be shown. The temperature of the elongate rod **22** will typically increase as the AC or DC current passes through it during the treatment procedure, but the urine (or saline) in the bladder **40** will prevent any significant increase in the temperature. The temperature of the elongate rod **22** may reach a temperature of around 104° F.

[0030] FIG. **5** shows the elongate rod **22** extending from the lumen **16**. In an embodiment, the distal end **26** may be rounded off smooth with approximately 2-3 mm of the distal end **26** being exposed from the end of the lumen **16**. The distal end **26** of the elongate rod **22** is spaced a distance from a biofilm **70** attached to an interior wall **72** of the bladder **40**. The AC or DC current is emitted from the distal end **26** and flows through the urine **38** which contains electrolytes and, therefore, is electrically conductive. Alternatively, the bladder **40** may be emptied just prior to the treatment procedure. In that circumstance, the doctor may fill the bladder **40** with saline via a separate catheter before turning on the AC or DC current in the elongate rod **22**. During treatment, the AC or DC current then flows through the biofilm **70** and disrupts or vaporizes the biofilm **70**. The AC or DC current then flows to the grounding pad **34** and back to the controller **32** via conductive wire **36**.

[0031] FIG. **6** shows another embodiment of the disclosure where the distal end **26** of the elongate rod **22** is position within the end of the lumen **16**. The end of the lumen **16** includes one or more holes **76** circumferentially arranged around the lumen **16**. The one or more holes **76** permit the AC or DC current to flow out of (escape) the distal end **26** of the elongate rod **22** and into the urine filled or partially filled bladder **400**.

[0032] In both FIGS. **5** and **6**, the distal end **26** of the elongate rod **22** is shown relatively close to biofilm **70** and the interior wall **72** of the bladder **40**. Because the urine **38** contains electrolytes and is, therefore, conductive, in general practice the distal end **26** may be positioned at or near the center of the interior of the bladder **40**. The urine (or alternatively, the saline) will conduct the AC or DC current to the biofilm **70** on all or nearly all of the interior wall **72** to disrupt the biofilm **70**.

[0033] FIGS. **7**A and **7**B shows a lumen **80** with the elongate rod **22** inserted in a central passageway **82** of the lumen **80**. The lumen **80** includes a side port **84** that extends parallel to central passageway **82**. As shown in FIG. **7**A, the lumen **80** is being pushed through a urethra **86**, but has not yet reached the interior of the bladder **40** and the urine **38** therein. As shown in FIG. **7**B, the end of the lumen **80** has entered the interior of the bladder **40** and is contacting the urine **38** therein. As explained above, the elongate rod **22** forms a water-tight seal with the central passageway **20** such that the urine **38** does not flow through the central passageway **20**. The side port **84**, however, adapted so that the urine **38** may flow through the side port **84** and out to the handle **18** of the lumen **16** so the doctor knows when the end of the lumen **16** has entered the bladder **40** and contacted urine **38**. The doctor may then insert the lumen **16** with more precision to position it in the central portion of the interior of the bladder **40** or closer to the biofilm **70** near the interior wall **72** of the bladder **40**.

[0034] In another embodiment, a very small thin strip of absorbent paper already in place along the elongate rod **22** which will turn dark when moistened by the urine **38**. The catheter **12** may also have a marking on its side to indicate the correct depth of placement of the lumen **16** into the bladder **40**.

[0035] As an alternative to urinary catheter **12** with the lumen **16** and the elongate rod **22** inserted within the lumen **16**, the treatment method may use an elongate rod coated with soft plastic or silicon except for an exposed distal end. Like the urinary catheter **12**, the exposed distal end of the coated elongate rod conducts AC or DC current into the fluid (urine or saline) in the bladder. Current flows from the controller **32**, to the elongate rod, to the urine **38** to the interior wall **72**, and finally to the grounding pad **34** placed externally on the abdomen.

[0036] The invention also contemplates a method for treating a patient with interstitial cystitis. The patient is instructed to arrive for the procedure with fairly full bladder. To begin the procedure in one embodiment, the patient lies on his or her back, i.e., supine, on the exam bed. In other embodiments, the procedure may begin with the patient lying on his or her back, i.e., prone, or the patient standing up. The doctor inserts the lumen **16** with the elongate rod **22** in the central passageway **20** through the patient's urethra **86** and into the desired position within the patient's bladder **40**. The grounding pad **34** is affixed to the skin of the patient in the desired location such as the patient's abdomen or suprapubis area. The conductive wires **30**, **36** are operatively coupled to the voltage (V) output of the controller **32**. Using the controller **32**, the doctor increases the power (AC or DC voltage) until the patient indicates they feel something and the power is held there for approximately 1-6 minutes, and in an embodiment 2-5 minutes, although shorter or longer times may be used depending on feedback from the patient. If needed, the power may be increased to the point where it begins to feel uncomfortable. In an embodiment, the AC or DC voltage applied to the elongate rod **22** may range between 2-50 volts, but higher voltages may also be applied. As the AC or DC current flows through the urine **38** (or saline or other suitable solution) in the bladder **40**, it necessarily has to pass through the biofilm **70** to reach the interior wall **72**, vaporizing the biofilm **70** on the interior wall **72** of the bladder **40**, which essentially eliminates the cause of the IC condition.

[0037] While the invention has been illustrated by a description of various embodiments, and while these embodiments have been described in considerable detail, it is not the intention of the Applicant to restrict or in any way limit the scope of the appended claims to such detail. Additional advantages and modifications will readily appear to those skilled in the art. The invention in its broader aspects is therefore not limited to the specific details, representative apparatus and method, and illustrative examples shown and described. Accordingly, departures may be made from such details without departing from the spirit or scope of the Applicant's general inventive concept.

What is claimed is:

1. A method for treating interstitial cystitis of a patient comprising:

generating a current at a first amperage in an interior of a bladder; and

applying the current for a predetermined time so as to disrupt a biofilm on or proximate to an interior wall of the bladder.

2. The method of claim **1**, wherein the predetermined time is 1-6 minutes.

3. The method of claim **2**, wherein the predetermined time is 2-5 minutes.

4. The method of claim **1**, further comprising:

after generating the current at the first amperage, increasing the current to a second amperage.

5. The method of claim **1**, further comprising:

placing a first end of an elongate rod inside an interior of the bladder; and

emitting the current from the first end of the elongate rod.

6. The method of claim **5**, further comprising:

operatively connecting the first end of the elongate rod to a controller; and

changing the current from the first amperage via the controller.

7. The method of claim **1**, wherein the elongate rod is disposed within a lumen routed through a urethra and into a bladder.

8. The method of claim **1**, wherein the patient is in a supine position during the treatment method.

9. The method of claim **1**, wherein the patient is in a prone position during the treatment method.

10. The method of claim **1**, wherein the patient is standing during the treatment method.

11. A method for treating interstitial cystitis of a patient comprising:

providing a catheter with a lumen and an elongate rod disposed within the lumen;

routing the lumen through a urethra such that a distal end of the elongate rod is disposed in an interior of a bladder;

connecting the elongate rod to a controller adapted to generate current;

supplying a current at a first amperage from the controller to the distal end of the elongate rod; and

applying the current for a predetermined time so as to disrupt a biofilm on or proximate to an interior wall of the bladder.

12. The method of claim **11**, wherein the predetermined time is 1-6 minutes.

13. The method of claim **12**, wherein the predetermined time is 2-5 minutes.

14. The method of claim **11**, further comprising:

after generating the current at the first amperage, increasing the current to a second amperage.

15. The method of claim **11**, wherein the patient is in a supine position during the treatment method.

16. The method of claim **11**, wherein the patient is in a prone position during the treatment method.

17. The method of claim **11**, wherein the patient is standing during the treatment method.

* * * * *